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To: U.S. Department of Health and Human Services (HHS) **Re:** Request for Information:
Accelerating the Adoption of Artificial Intelligence in Clinical Care **Subject:** Technical Advisory
on Data Infrastructure and Hardware Drift

To the Committee:

I am writing to provide technical feedback on the infrastructure requirements necessary for the safe acceleration of AI in clinical settings. As a Data Engineering Manager currently overseeing petabyte-scale genomic pipelines in the life sciences industry (Corteva Agriscience), I manage the "Data Gravity" and lifecycle challenges of biological AI daily.

While the RFI correctly identifies the potential of AI, it currently overlooks a fundamental engineering failure mode: **Silent Instrument Drift**.

1. The Engineering Gap: "Batch Effects" in Production In industrial genomics, we frequently observe that AI models are highly sensitive to the specific firmware and calibration of the data capture hardware (e.g., DNA sequencers or imaging arrays). A model trained on data from "Sequencer A" will often degrade silently when applied to "Sequencer B," interpreting the hardware noise profile as a biological signal.

In a clinical setting, if a hospital upgrades its MRI firmware or sequencing hardware, an "accelerated" AI model lacking rigid provenance checks will likely produce diagnostic errors that are statistically undetectable until patient harm occurs.

2. Recommendation: Mandate Instrument-Agnostic Validation To safely accelerate adoption, HHS standards must look upstream of the model layer. I recommend that any framework for AI adoption include a mandatory **"Data Engineering Competency"** check:

- **Metadata Provenance:** AI systems must be required to ingest and validate the *hardware metadata* (Device ID, Firmware Version) of the input data.
- **Drift Normalization:** Protocols must be established to distinguish between *Biological Drift* (the patient changing) and *Instrument Drift* (the machine changing).

We have solved this in the private sector by enforcing strict schema contracts on our data lakes. Clinical systems require the same rigor. Adopting AI without standardizing the underlying data engineering layer will result in fragile systems that fail at scale.

Respectfully,

Anil Bodepudi

